

ORLAND, Norway

WMO: 012410

Lat: 63.699N

Lon: 9.604E

Elev: 28

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.9	17.8	5.8	7.4	17.9	9.3	8.9	21.2	42.8	37.0	39.1	36.4	14.7	130	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	74.6	61.8	70.8	59.8	67.5	58.2	63.3	71.6	61.1	68.2	59.3	65.3	10.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	76.1	66.5	57.8	71.5	63.9	56.4	67.8	61.9	28.7	71.5	27.2	68.2	25.9	65.5	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.9	31.4	27.7	DB	8.7	80.6	4.3	4.4	5.6	83.8	3.1	86.3	0.6	88.8	-2.5	92.0	(4)
(5)			WB	7.8	66.8	4.3	3.0	4.7	68.9	2.1	70.7	-0.3	72.4	-3.4	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	33.2	32.8	35.1	41.6	47.6	52.5	57.3	57.4	52.7	44.8	38.1	34.7
	DBStd	10.82	6.48	7.15	6.16	5.72	6.21	5.28	5.38	5.11	5.01	5.56	6.48	7.31
	HDD50	2912	521	481	462	259	122	28	1	2	28	176	357	475
	HDD65	7663	986	901	927	701	539	376	248	242	370	627	806	940
	CDD50	746	0	0	0	8	49	104	229	232	109	14	1	0
	CDD65	20	0	0	0	0	1	1	10	8	0	0	0	0
	CDH74	129	0	0	0	0	4	11	73	41	0	0	0	0
	CDH80	16	0	0	0	0	0	0	12	4	0	0	0	0
Wind	WSAvg	13.1	17.7	16.9	15.2	12.0	10.7	9.9	8.4	8.9	11.9	14.0	15.4	16.6
Precipitation	PrecAvg	39.9	3.3	3.0	2.7	2.2	2.0	2.6	2.9	3.4	4.7	4.6	3.7	4.4
	PrecMax	58.5	8.8	8.1	6.9	5.0	4.0	7.6	5.4	7.0	12.2	9.5	8.7	10.8
	PrecMin	24.3	0.1	0.2	0.4	0.6	0.6	0.6	0.5	0.9	1.1	0.9	0.2	1.3
	PrecStd	7.0	2.0	1.6	1.5	1.1	0.8	1.3	1.3	1.5	2.2	2.1	2.0	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	46.8	51.2	63.5	73.2	75.5	81.5	78.6	69.1	60.8	53.4	49.7
		MCWB	42.6	42.6	42.8	50.7	57.5	59.2	66.1	63.6	58.0	53.2	48.0	46.4
	2%	DB	44.7	44.9	46.9	57.5	67.2	69.9	75.0	73.7	65.5	57.2	50.2	47.3
		MCWB	41.8	41.6	41.3	47.4	54.5	57.2	62.9	61.8	56.4	50.9	46.0	44.4
	5%	DB	43.0	43.2	44.9	53.4	62.0	65.7	70.8	69.8	62.5	54.2	48.2	45.3
		MCWB	40.0	40.2	40.5	45.6	51.9	56.0	61.1	60.1	55.2	49.2	44.8	42.5
	10%	DB	41.3	41.5	43.1	49.8	57.6	62.0	66.9	66.2	59.9	52.0	46.5	43.5
		MCWB	38.5	38.5	39.6	43.6	49.5	54.6	59.1	58.3	53.9	47.7	42.8	40.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	44.6	44.8	51.8	58.5	61.3	68.2	65.3	59.9	54.3	49.9	47.6
		MCDB	45.2	45.6	47.3	61.3	70.4	70.1	79.4	75.5	65.8	59.0	52.0	48.7
	2%	WB	42.6	42.8	43.0	48.4	55.2	58.5	64.0	63.2	57.6	51.9	47.4	45.2
		MCDB	44.0	44.1	45.4	55.9	64.4	66.7	72.6	71.5	62.5	55.6	49.1	46.4
	5%	WB	40.6	40.8	41.4	46.3	52.6	56.6	61.8	61.0	56.2	50.2	45.3	43.0
		MCDB	42.2	42.7	44.2	51.2	60.6	63.5	69.2	67.7	60.7	53.3	47.6	44.7
	10%	WB	38.9	38.8	39.9	44.5	50.1	54.7	59.8	59.2	54.9	48.6	43.3	41.0
		MCDB	40.9	41.0	42.7	48.4	55.7	60.7	66.0	64.6	59.1	51.5	45.7	43.1
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.4	8.9	11.0	11.9	10.8	12.0	12.2	10.3	8.0	6.9	7.1
		MCDBR	7.0	7.8	10.1	16.2	19.7	18.7	20.6	19.2	13.9	9.7	7.9	7.9
		MCWBR	6.5	6.7	6.8	8.5	9.7	9.6	10.7	9.2	7.1	6.2	6.5	7.3
	5% WB	MCDBR	6.9	7.3	9.2	14.1	18.0	16.7	19.0	17.0	12.3	8.8	7.9	7.8
		MCWBR	6.9	7.0	7.0	8.0	9.6	9.5	10.8	9.1	7.2	6.5	7.3	7.6
Clear-Sky Solar Irradiance	taub	N/A	0.306	0.333	0.356	0.354	0.356	0.364	0.352	0.337	0.312	0.363	0.359	
	taud	N/A	2.232	2.304	2.369	2.441	2.468	2.472	2.502	2.522	2.416	2.598	2.514	
	Ebn at Noon	N/A	193	239	261	273	274	268	261	242	195	104	51	
	Edh at Noon	N/A	19	26	30	31	31	30	27	22	16	8	5	
All-Sky Solar Radiation	RadAvg	41	191	577	1104	1472	1585	1476	1132	669	291	69	19	
	RadStd	7	32	83	91	77	153	133	116	81	42	13	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.26	N/A	N/A	N/A	N/A	N/A	-306	-452	+151	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-257	-340	+121	N/A

Nomenclature: See separate page